MULTIVARIATE ASSESSMENT OF STUDENT PERFORMANCE

POLITECNICO MILANO 1863

GROUP 9
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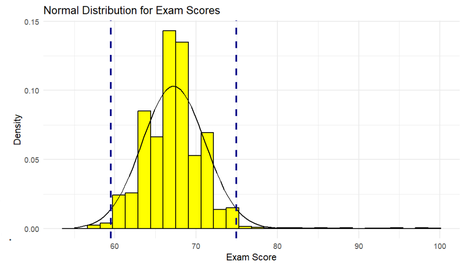
OBJECTIVE

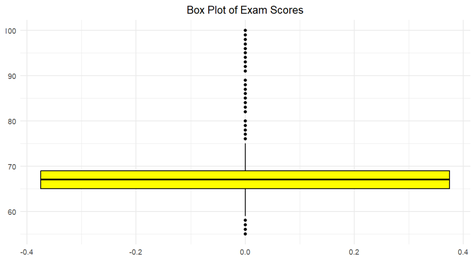
Analyzing the determinants influencing students' overall exam scores utilizing the provided dataset. The study employs various statistical analysis models learned throughout the course. The objective is to identify key factors significantly impacting students' performance and provide valuable insights based on data-driven findings.

EXPLORATORY DATA ANALYSIS

For the modelling of the dataset, we decided to use 'EXAM_SCORES' as our dependent variable and the rest of the variables as independent variables. A collinearity check was done in order to check for multicollinearity after the categorical variables were changed into respective dummies as numerical variables. Interestingly most variables showed extremely negligible values of negative collinearity. Next we looked at the normal distribution of our dependent variable to check for outliers. The normal distribution curve showed the presence of comparatively high number of outliers. These outliers were students who scored extremely well in the exam and hence moved far away for the mean. For the sake of the objective these outliers cannot be simply removed, so we decided to divide the dataset into two - Student data low (Students who scored between 55-79) and Student data high (studentswho scored between 80-100). The models analysed these two groups separately and provided insights.

```
'data.frame':   6606 obs. of  20 variables:
 $ Hours_Studied             : int  18 23 14 28 28 16 15 21 25 18 ...
 $ Attendance                : int  89 83 90 90 96 83 83 74 99 93 ...
 $ Parental_Involvement      : chr  "High" "High" "High" "Low" ...
 $ Access_to_Resources       : chr  "Medium" "High" "High" "Medium" ...
 $ Extracurricular_Activities: chr  "Yes" "Yes" "Yes" "Yes" ...
 $ Sleep_Hours               : int  4 4 8 9 4 8 7 6 7 7 ...
 $ Previous_Scores           : int  73 89 86 91 98 92 97 94 77 76 ...
 $ Motivation_Level          : chr  "Medium" "Low" "Medium" "Medium" ...
 $ Internet_Access           : chr  "Yes" "Yes" "Yes" "Yes" ...
 $ Tutoring_Sessions         : int  3 1 4 0 1 2 2 1 0 2 ...
 $ Family_Income             : chr  "High" "Medium" "Medium" "Medium" ...
 $ Teacher_Quality           : chr  "Medium" "Medium" "Medium" "Medium" ...
 $ School_Type               : chr  "Private" "Public" "Private" "Public" ...
 $ Peer_Influence            : chr  "Positive" "Negative" "Negative" "Positive" ...
 $ Physical_Activity         : int  2 3 2 2 3 4 2 1 3 4 ...
 $ Learning_Disabilities     : chr  "No" "No" "No" "No" ...
 $ Parental_Education_Level  : chr  "College" "High School" "High School" "College" .
 $ Distance_from_Home        : chr  "Near" "Far" "Near" "Moderate" ...
 $ Gender                    : chr  "Female" "Male" "Female" "Female" ...
 $ Exam_Score                : int  100 99 99 98 98 98 97 97 97 96 ...
```



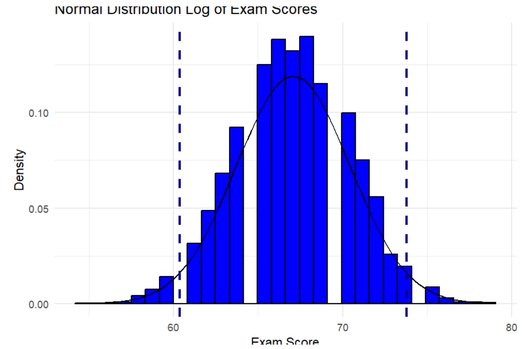


STUDENT PERFORMANCE FACTOR DATA → STUDENT DATA LOW / STUDENT DATA HIGH

DATA TRANSFORMATION

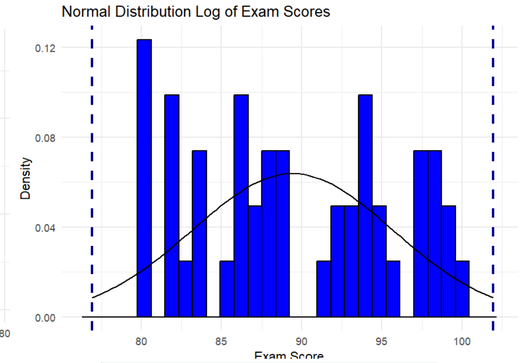
Log transformation of the Dependent variable in both the datasets

STUDENT DATA LOW



6559 obs. of 20 variables

STUDENT DATA HIGH



47 obs. of 20 variables

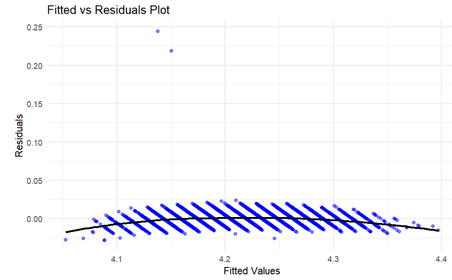
LINEAR MIXED MODEL WITH RANDOM EFFECT

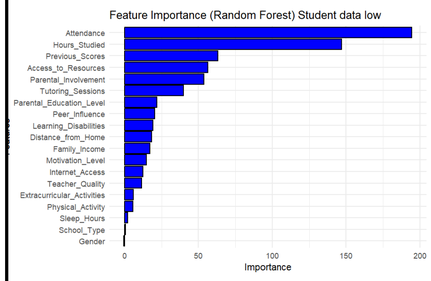
After plenty of iterations a Linear Mixed model with random effect and interactions was selected as the final model. The variable 'Parental_Involvement' was selected as the random effect.

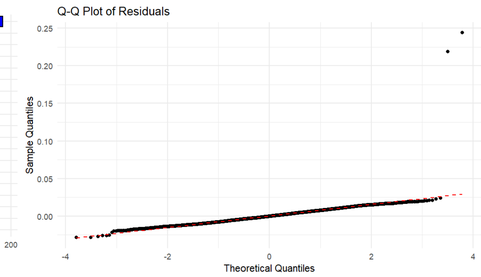
$$\log(Exam\ Score)_i = \beta_0 + \beta_1 * Attendance_i + \beta_2 * Hours\ studied_i + \beta_3 * Previous\ Scores_i + \beta_4 * Parental\ Involvement_i + \beta_5 * Access\ to\ Resources_i + \beta_6 * Distance\ from\ Home_i + \beta_7 * Learning\ Disabilities_i + \beta_8 * Physical\ Activities_i + \gamma_1 * (Teacher\ Quality_i * Learning\ Disabilities_i) + \gamma_2 * (Distance\ from\ Home_i * Learning\ Disabilities_i) + b_i + \varepsilon_i$$

STUDENT DATA LOW

```
Fixed effects:
                                   Estimate Std. Error t value
(Intercept)                       3.716e+00  3.815e-03 974.173
Attendance                        2.936e-03  9.065e-06 323.875
Hours_Studied                     4.387e-03  1.747e-05 251.114
Motivation_Level                  7.538e-03  1.504e-04  50.133
Internet_Access                   1.494e-02  3.962e-04  37.713
Access_to_Resources               1.460e-02  1.498e-04  97.448
Sleep_Hours                       1.465e-04  7.123e-05   2.056
Tutoring_Sessions                 7.295e-03  8.500e-05  85.824
Previous_Scores                   7.178e-04  7.277e-06  98.644
Gender                            7.641e-05  2.116e-04   0.361
Parental_Involvement              1.471e-02  1.506e-04  97.667
Extracurricular_Activities        7.280e-03  2.131e-04  34.166
Peer_InfluenceNeutral             7.472e-03  2.835e-04  26.358
Peer_InfluencePositive            1.489e-02  2.825e-04  52.696
Teacher_Quality                   7.784e-03  3.227e-04  24.122
School_Type                       2.012e-03  8.692e-04   2.315
Learning_Disabilities            -1.818e-02  1.768e-03 -10.282
Distance_from_Home               -7.483e-03  1.651e-04 -45.318
Physical_Activity                 3.372e-03  1.077e-04  31.320
Teacher_Quality:School_Type      -8.651e-04  3.814e-04  -2.268
Teacher_Quality:Learning_Disabilities  2.061e-03  5.767e-04   3.575
Learning_Disabilities:Distance_from_Home  1.480e-03  5.058e-04   2.927
Learning_Disabilities:Physical_Activity  -8.622e-04  3.215e-04  -2.681
```







AIC	BIC	LogLik	Conditional R Square	BP test	p Value
-43927	-43758	2422.3	0.971	7.2916	0.006928

STUDENT DATA HIGH

```
Fixed effects:
                                   Estimate Std. Error t value
(Intercept)                       3.9983188  0.1079297  37.046
Attendance                        0.0027131  0.0007702   3.523
Hours_Studied                     0.0033397  0.0014293   2.337
Motivation_Level                 -0.0023050  0.0097553  -0.236
Internet_Access                  -0.0106518  0.0282755  -0.377
Access_to_Resources               0.0404414  0.0100334   4.031
Sleep_Hours                      -0.0057432  0.0042579  -1.349
Tutoring_Sessions                 0.0016094  0.0064970   0.248
Previous_Scores                   0.0008837  0.0004252   2.078
Gender                            0.0125993  0.0144878   0.870
Parental_Involvement              0.0059078  0.0103644   0.570
Extracurricular_Activities       -0.0062219  0.0152727  -0.407
Peer_InfluenceNeutral            -0.0185132  0.0200594  -0.923
Peer_InfluencePositive            0.0131753  0.0204880   0.643
Teacher_Quality                   0.0547765  0.0284910   1.923
School_Type                       0.0864043  0.0753205   1.148
Learning_Disabilities             0.4461065  0.3485679   1.280
Distance_from_Home               -0.0092193  0.0104880  -0.879
Physical_Activity                 0.0122125  0.0064696   1.888
Teacher_Quality:School_Type      -0.0537794  0.0333370  -1.613
Teacher_Quality:Learning_Disabilities  -0.1670299  0.1190434  -1.403
Learning_Disabilities:Distance_from_Home  -0.0628484  0.0873965  -0.719
Learning_Disabilities:Physical_Activity  0.0253984  0.0324758   0.782
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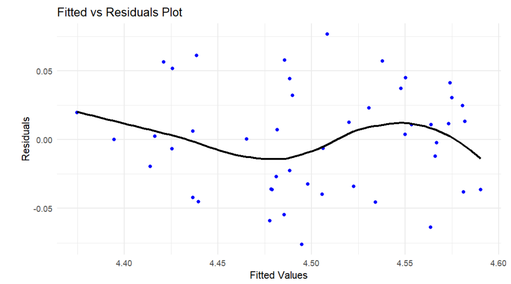

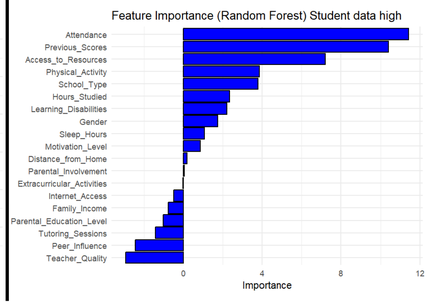

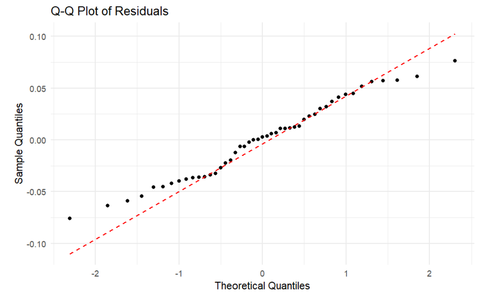

AIC	BIC	LogLik	Conditional R Square	BP test	p Value
-125	-78.71	87.48	0.703	0.1155	0.7339

KEY DETERMINANTS OF STUDENT PERFORMANCE

COMMON POSITIVE INFLUENCES ON STUDENT PERFORMANCE

Our logistic regression analysis on both the Student Data Low and Student Data High datasets identifies several common factors that positively impact student performance:

- **Attendance:** Regular class attendance is strongly associated with better academic outcomes.
- **Hours Studied**: The amount of time dedicated to studying positively correlates with improved performance.
- **Parental Involvement**: Higher engagement from parents contributes significantly to student success.
- **Teacher Quality**: The effectiveness of teachers plays a crucial role in shaping student achievement.

Additionally, an interesting **interaction effect** emerged between **Teacher Quality and Learning Disabilities.** This suggests that students with learning disabilities tend to perform better when they receive additional attention and tailored teaching methods from high-quality educators.

COMMON NEGATIVE INFLUENCES ON STUDENT PERFORMANCE

Conversely, the following factors were consistently associated with poorer exam performance:

- **Distance from Home**: Students who live farther from school tend to experience lower academic performance, potentially due to longer commute times and fatigue.
- **Learning Disabilities**: While targeted interventions can mitigate challenges, learning disabilities remain a significant factor influencing academic performance.

KEY DIFFERENTIATORS BETWEEN AVERAGE AND HIGH-PERFORMING STUDENTS

When comparing students who perform at an average level with those in the 80th percentile, the distinguishing factors are:

- **Physical Activity**: Students who engage in regular physical exercise tend to achieve higher academic performance, possibly due to improved cognitive function and reduced stress levels.
- **Previous Scores**: Past academic performance strongly predicts future success, reinforcing the idea of cumulative learning effects.
- **Access to Resources:** Better access to educational materials, tutoring, and digital learning tools enhances student outcomes.

These findings indicate two key insights:

1. **Academic trajectories tend to be self-reinforcing, meaning students who have historically performed well are likely to continue excelling.**
2. **Improved access to resources and physical activity can significantly enhance academic performance, even for students who previously scored at an average level.**

IMPACT OF PARENTAL EDUCATION LEVEL

Incorporating Parental Education Level as a random effect improves the model's ability to explain performance variance, as evidenced by a better R^2 value. This suggests that students with more highly educated parents tend to benefit from additional academic support at home.

STATISTICAL TESTS AND MODEL VALIDATION

1. Likelihood Ratio Test

The Likelihood Ratio Test (LRT) results, as summarized in Table 1, guided the selection of the optimal mixed-effects model, helping to identify the most significant interaction terms and random effect variables.

2. Breusch-Pagan Test

The Breusch-Pagan test revealed the presence of heteroskedasticity in the model, suggesting that the variance of errors is not constant. To further validate our results, we employed a Random Forest Algorithm as a non-parametric alternative.

- The Random Forest model largely corroborated the logistic regression findings for the Student Data Low dataset.
- However, for the Student Data High dataset, there were some deviations, particularly in identifying the negative predictors of student performance. This discrepancy indicates that different statistical approaches may capture distinct underlying patterns within high-performing student groups.

3. AIC and BIC Test

To complement our model selection process, we examined the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). These measures provided additional insights into the trade-off between model complexity and explanatory power. Lower AIC and BIC values indicated a more efficient model fit, reinforcing our selected logistic regression structure. The alignment of AIC/BIC results with our likelihood ratio and Breusch-Pagan tests further strengthens the validity of our findings.

CONCLUSION

Our findings highlight the importance of attendance, study habits, parental involvement, and teacher quality in driving student success. Additionally, access to resources and physical activity play a key role in distinguishing top performers. While our primary model offers valuable insights, validation through alternative techniques like Random Forest ensures robustness in our conclusions